

SentinellQ - Ensemble Anomaly Detection Report

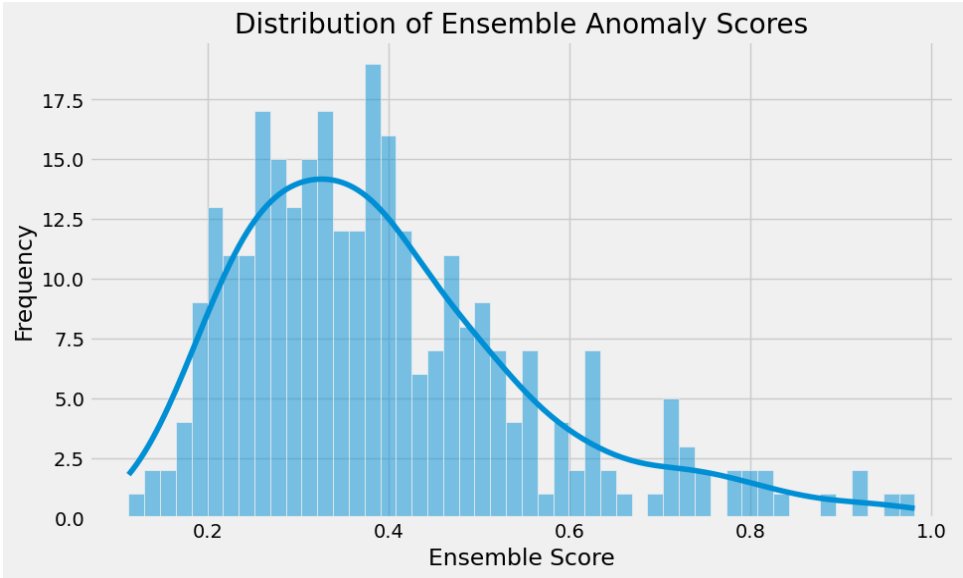
Executive Summary

This report details the performance and prioritization of the final ensemble anomaly detection model. The ensemble combines Isolation Forest, One-Class SVM, EWMA, CUSUM, Persistence Score, and Incident Probability to provide a robust anomaly detection signal. Anomalies are prioritized based on their ensemble scores.

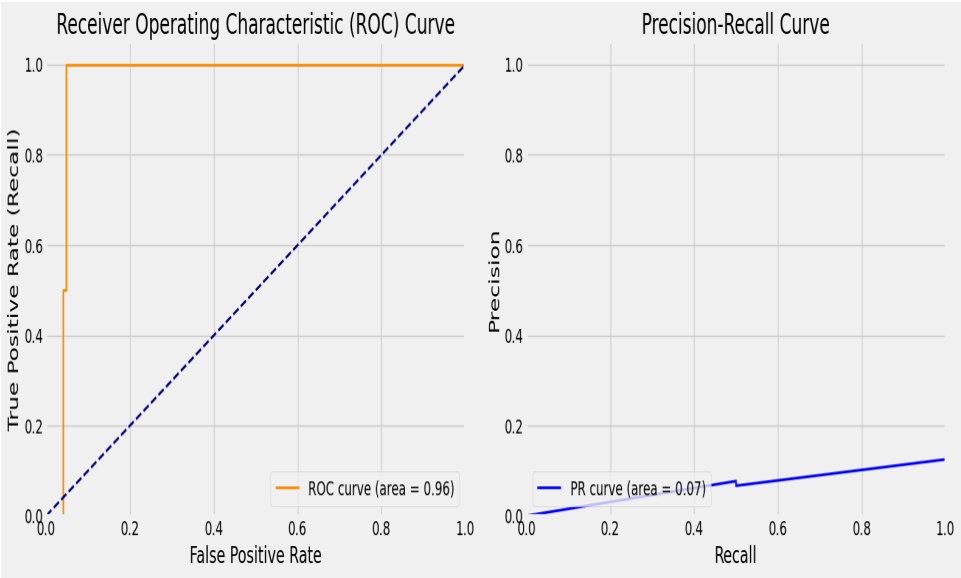
Ensemble Model Evaluation

Metric	Value
Model	Final Ensemble
Precision	1.0000
Recall	0.0279
F1	0.0542
ROC_AUC	0.9995
PR_AUC	0.9913

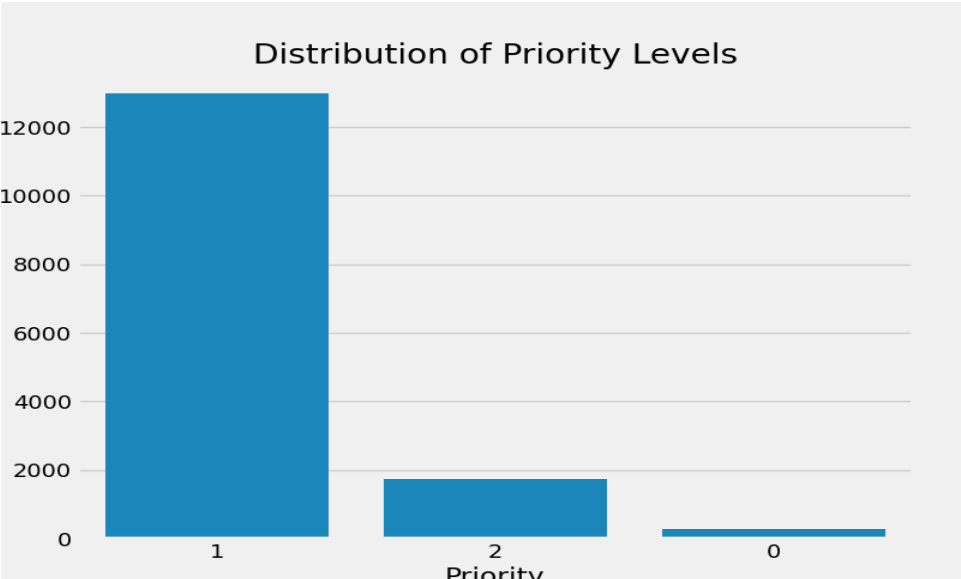
Distribution of Final Ensemble Anomaly Scores



ROC and Precision-Recall Curves



Distribution of Anomaly Priority Levels



Top 10 Highest Scoring Anomalies

Timestamp	Service	Latency	Score	Priority
2026-01-04 17:36:00	fraud-detection	417	0.9020042899779401	Critical
2026-01-04 17:46:30	fraud-detection	307	0.8584823798179708	Critical
2026-01-01 14:53:30	notification-service	107	0.7825728996496273	Critical
2026-01-04 17:25:00	notification-service	120	0.7698892348128507	Critical

2026-01-01 14:43:30	notification-service	120	0.7473891836013900	Critical
2026-01-04 17:34:00	auth-service	134	0.7231471862267245	Critical
2026-01-04 17:34:30	notification-service	157	0.7206394633933559	Critical
2026-01-01 14:25:00	notification-service	120	0.7194413706054587	Critical
2026-01-04 17:35:00	trading-engine	187	0.7182143076918222	Critical
2026-01-01 14:05:00	portfolio-service	170	0.7154624620707093	Critical